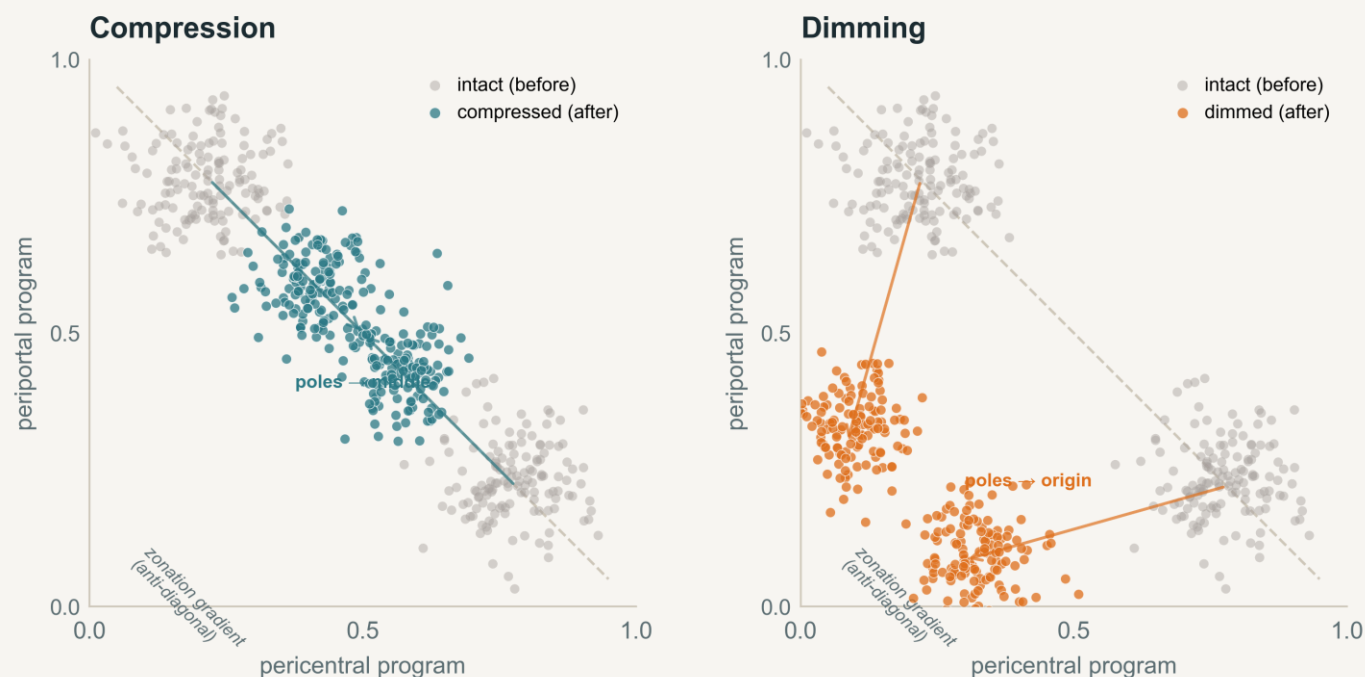


Beyond the boxes — zonation is a gradient

Zonation is a continuous gradient — cells strung on an anti-diagonal. It can weaken two ways **without any cell changing class**:

Two de-zonation modes in anchor-program space · SCHEMATIC (illustrative, synthetic cells)



Compression — shape

Balance drifts to the middle; the poles' contrast shrinks.

Mild, non-monotone → gradient present.

Dimming — level

The pericentral detox program declines in level — same balance, lower volume.

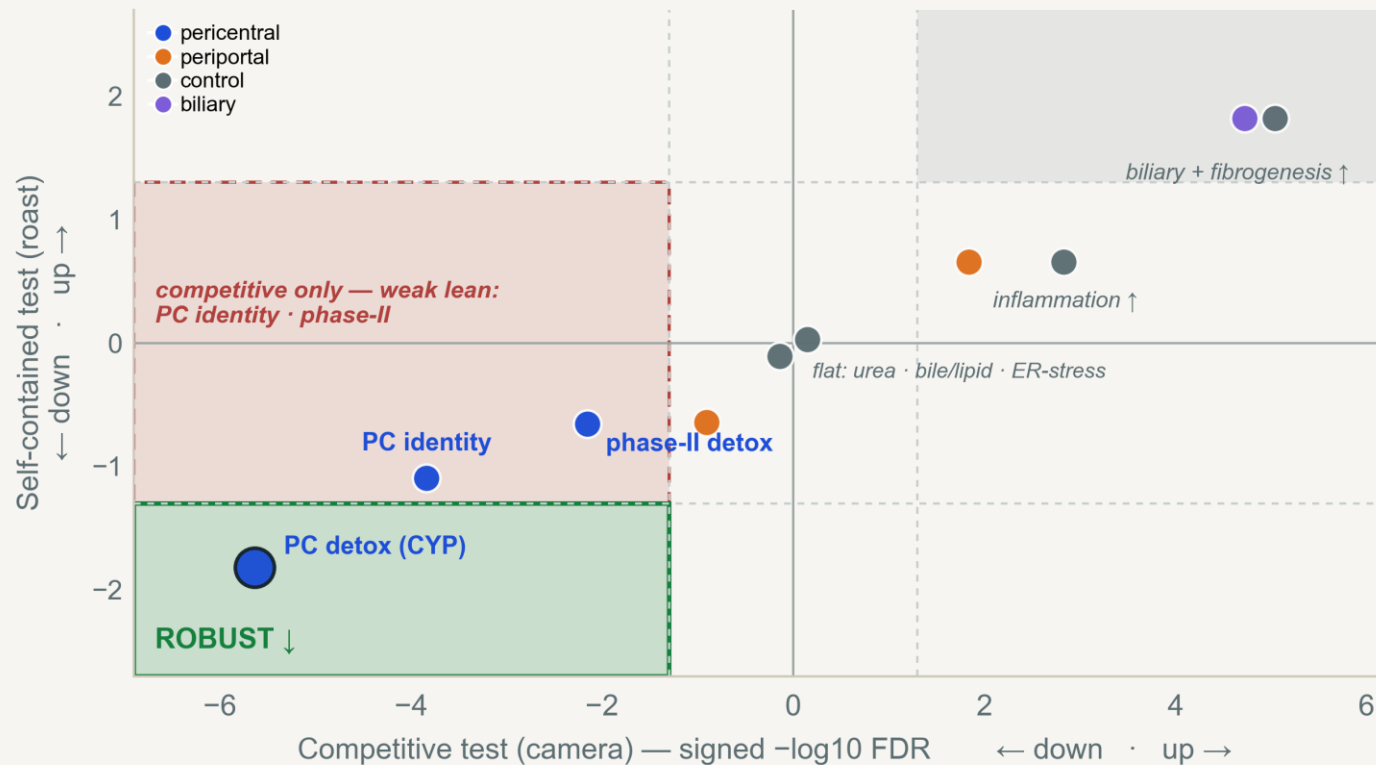
Real coordinated decline → a finding (Results).

Matched biopsies: shape holds (no compression); level dims (detox — confirmed next). *Only the explant collapses to one pole.*

Only one program robustly changes — pericentral detox

Each dot is one pre-specified program. Right = it moves more than the genome (competitive); up/down = it moves on its own (self-contained). Only a dot in a far corner is robust.

Of every program tested, only pericentral detox is robust in both tests



Pericentral detox (CYP)

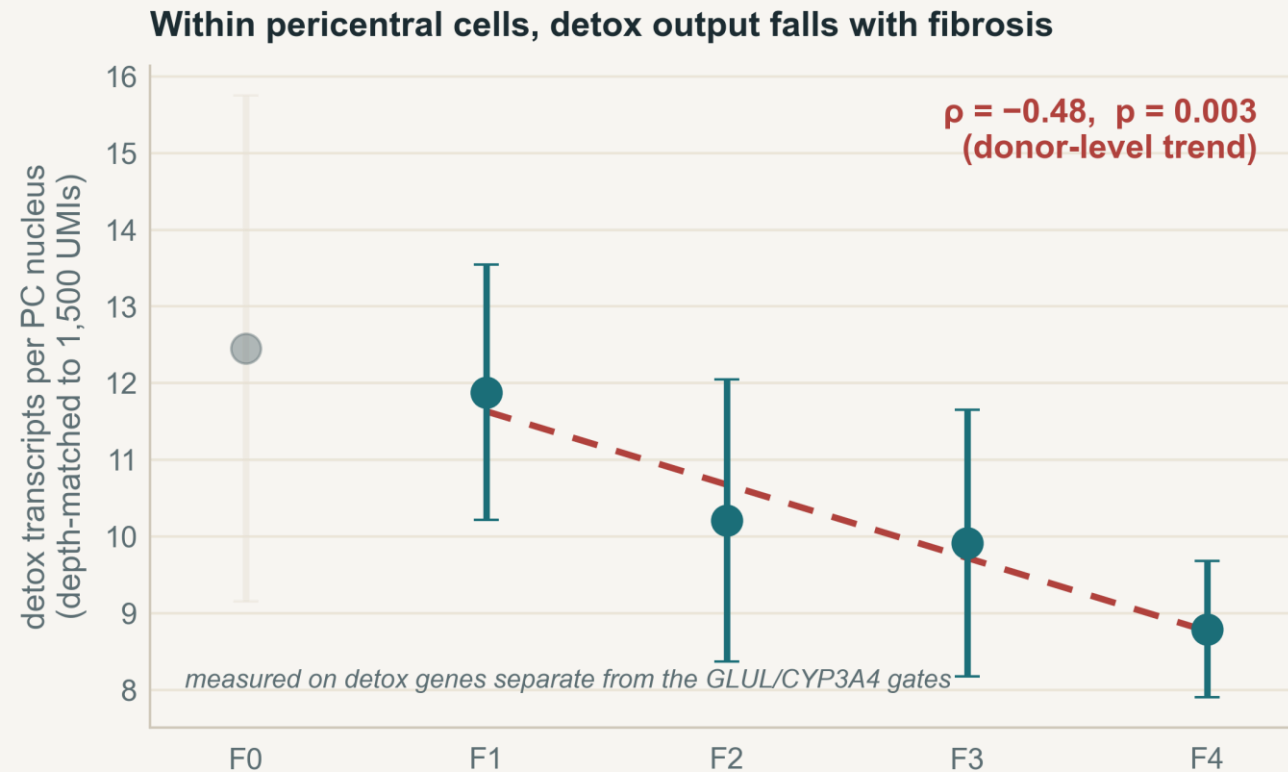
falls coherently and hard enough → clears both tests + the per-cell trend ($p=-0.48$). The one robust change.

WHY THE OTHERS AREN'T ROBUST

The landmark & phase-II sets only lean down **faintly** — far too weakly to clear the self-contained test (no single gene moves). **A shift this faint isn't a real turn-off**; only the CYP detox core moves hard enough to count as a robust change.

Controls behaved ↑ — the test reads real biology.

Pericentral detox dims — dimming, not de-zonation



Per-cell, depth-matched

composition can't fake it — the decisive evidence ($\rho = -0.48$).

Cells keep their class

the pericentral anchor fractions stay flat — no de-zonation; no identity gene is individually lost.

Coordinated, not a switch

no single gene clears FDR — many genes each shift a little, same direction.

Like a radio: same genre (cells stay pericentral), lower volume (the detox channel dims).

A relative decline within a fixed budget, not proven absolute molecule loss; a donor-level trend; could partly reflect a shift among PC subzones.

■ CONCLUSION

No de-zonation signal — the pericentral detox program dims

- 01** No transcriptional de-zonation signal — cells keep their zonal class (large shift excluded)
- 02** The one real change: the pericentral detox program dims (gates held)
- 03** Much of the dramatic trajectory may track acquisition, not disease

We correct one evidence leg — the snRNA marker-correlation; the paper's imaging and protein staining are untouched. The biliary signal is a separate compositional lead.

Scope — a “PC-like program,” not lobular location; a weak coordinated gene-set program is not excluded; F4 is only 4 donors.